6.3 Electromagnetism

This section provides knowledge and understanding of magnetic fields, motion of charged particles in magnetic fields, Lenz's law and Faraday's law. The application of Faraday's law may be used to demonstrate how science has benefited society with important devices such as generators and transformers. Transformers are used in the transmission of electrical energy using the national grid and are an integral part of many electrical devices in our homes. The application of Lenz's law allows discussion of the use of scientific knowledge to present a scientific argument (HSW1, 2, 3, 5, 6, 7, 8, 9, 11, 12).

6.3.1 Magnetic fields

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	magnetic fields are due to moving charges or permanent magnets	
(b)	magnetic field lines to map magnetic fields	
(c)	magnetic field patterns for a long straight current-carrying conductor, a flat coil and a long solenoid	
(d)	Fleming's left-hand rule	HSW7
(e)	**(i)** force on a current-carrying conductor;	

(ii) techniques and procedures used to determine the uniform magnetic flux density between the poles of a magnet using a current-carrying wire and digital balance

(f) magnetic flux density; the unit tesla.